



DATA-548 PROJECT REPORT

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Abstract:

Mental health has become an important topic and more openly discussed. This project explores a Kaggle dataset of mental health statements categorized as Normal, Stress, Anxiety, Suicidal, Personality Disorder, Depression, and Bipolar mental health statuses. Through an exploratory data analysis, patterns of class distribution, text lengths, and key words were identified in each mental health category. By identifying these patterns, we were able to create a model to accurately predict the mental health statuses attached to each statement.

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Introduction

Globally, mental health is a growing concern in the United States and around the world. It is estimated that one in five U.S. adults experience mental illness each year and one in six U.S. youth, ages 6-17, experience mental health disorders each year². This is especially concerning since Suicide is the 2nd leading cause of death among people ages 10-14². These are preventable deaths that should never happen.

The primary goal of the analysis is to familiarize ourselves with the various mental health classes and to identify patterns. Our objective is to visualize the distribution of the mental health statuses, evaluate the length of the statements per class, explore frequently used words through word clouds, and accurately predict the mental health statuses for each statement.

Data Description

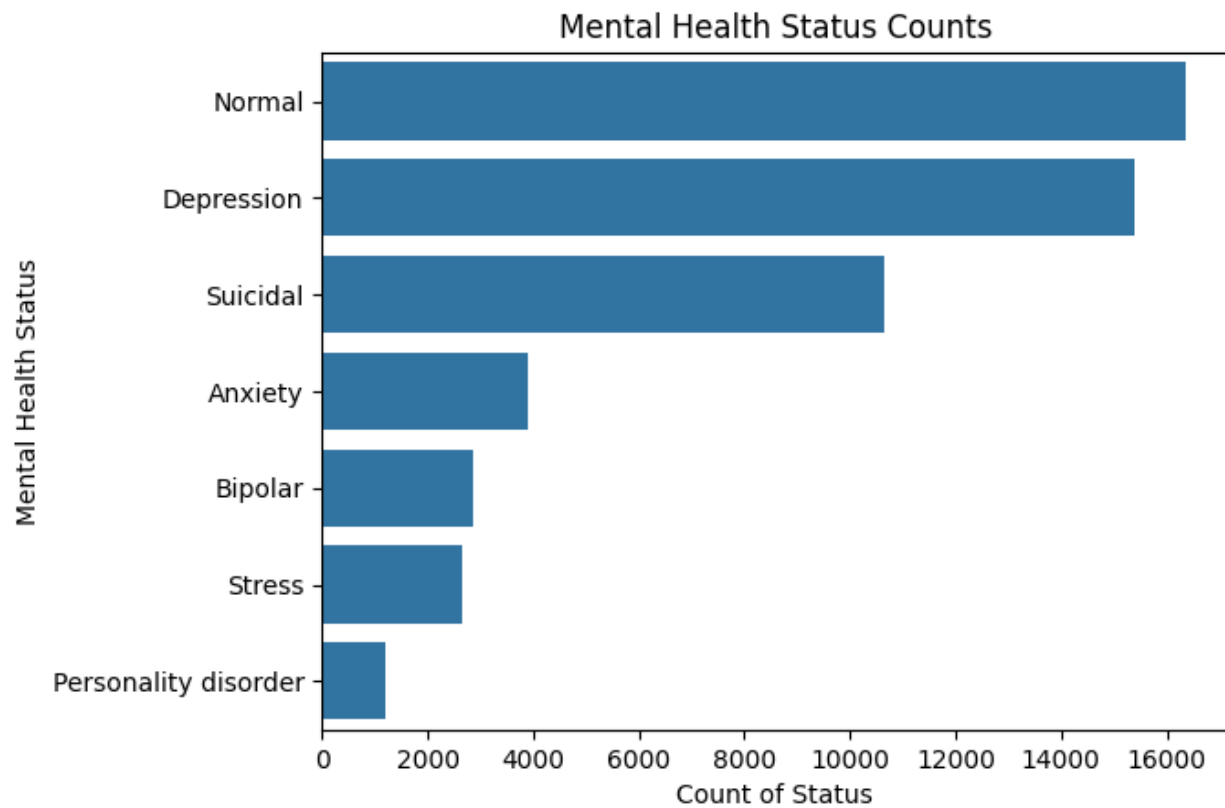
This dataset was found on Kaggle¹. The data is a collection of mental health statuses tagged from various statements collected from many other mental health datasets on Kaggle. It contains 53,042 rows and three columns. Each row contains a mental health statement where each row represents a unique individual comment. There are no missing values in this dataset.

Column name	Type	Description
Unique_id	int	Unique identifier for each statement.
Statement	String	Mental health statement.
Status	String	Mental health status (Normal, Stress, Anxiety, Suicidal, Personality disorder, Depression, Bipolar)

Exploratory Data Analysis

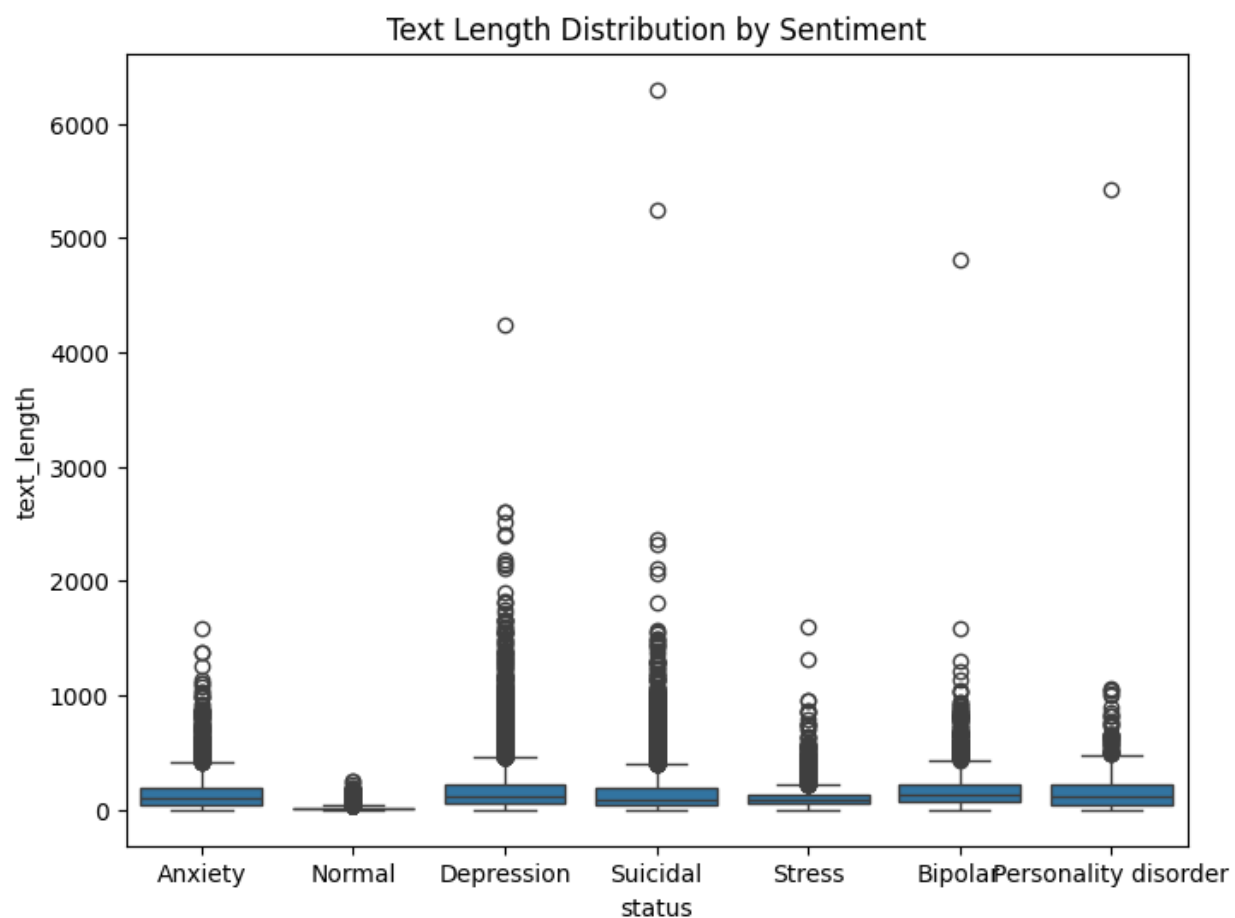
To get a better understanding of the dataset we created multiple plots and visualizations to find any trends/correlations between the statements and mental health status. The goal of this analysis is to familiarize ourselves with the class distributions and text features to see if any cleaning is necessary.

Figure 1



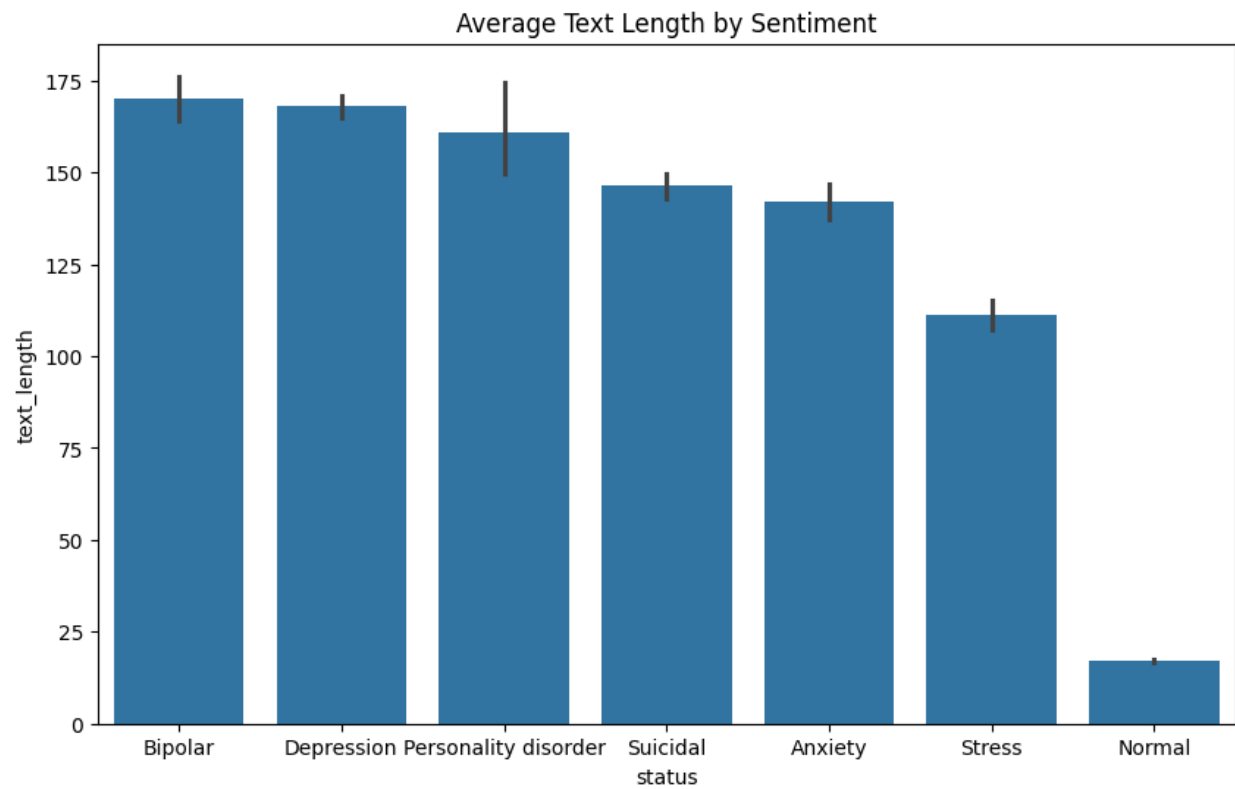
From the count plot above we can see a majority of cases are Normal, Depression, and Suicidal.

Figure 2

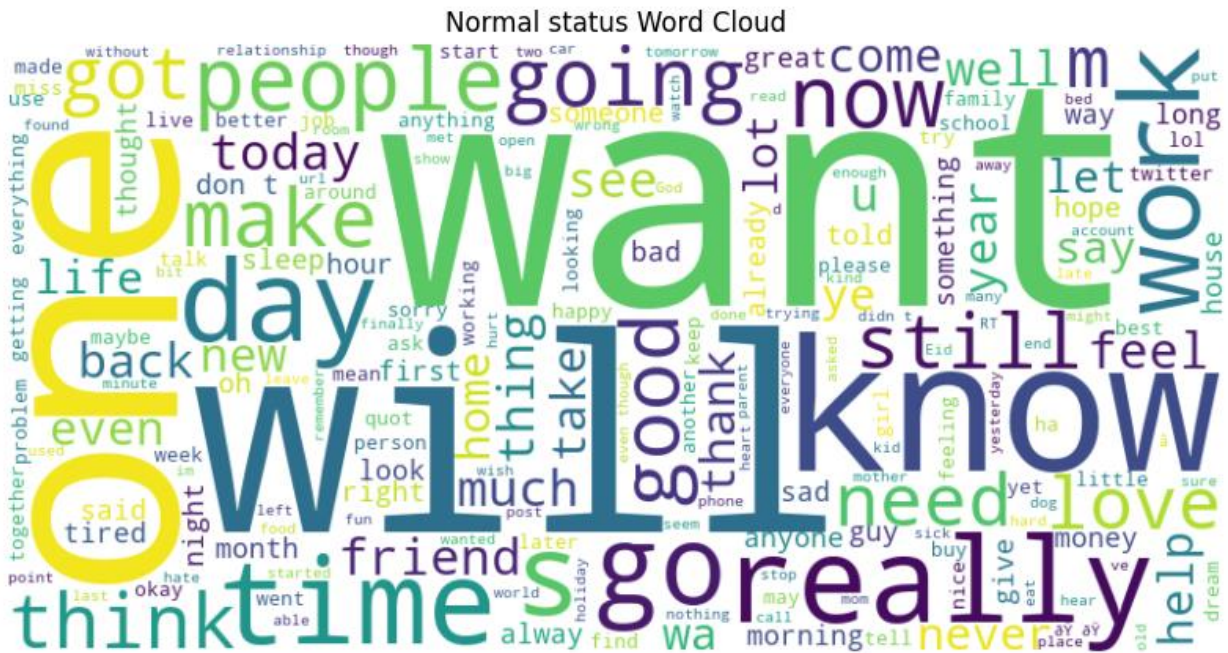


From the above chart we can see there are some outlier statements with exceptionally long text lengths. The majority of statements are below 3,000. We can also see the Normal class has the shortest text lengths overall.

Figure 3

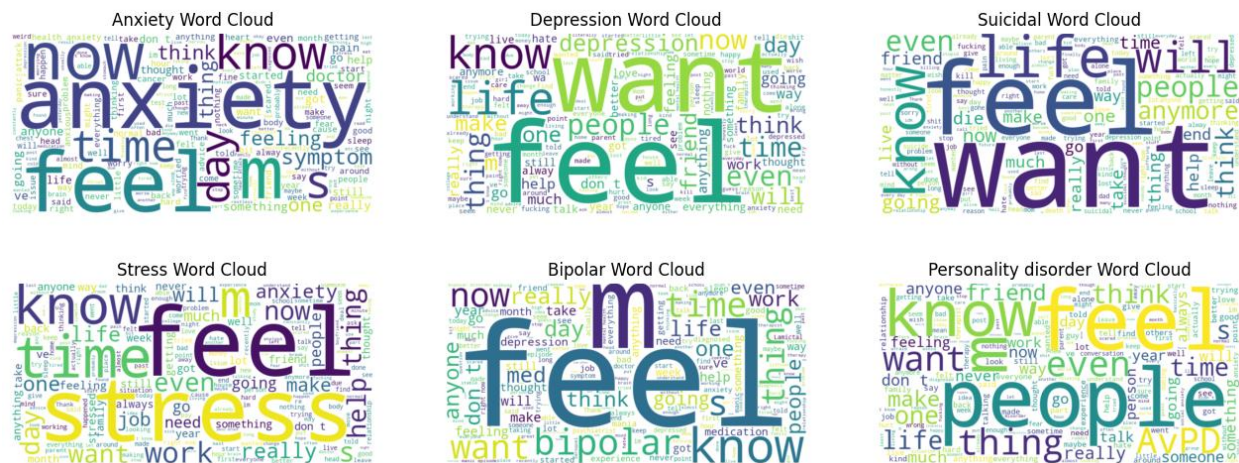


From the above chart we can see the statements classified as Normal have the least average text length, while statements with mental health problems tend to have longer statements on average.



Nothing really stands out to me as an indicator of the Normal class in the above word cloud visualization.

Figure 5



The above word clouds do not give me too much insight. The main thing I notice from all the classes with mental health issues is "feel", "know", and the words of their classification showing up in the clouds.

Modeling

For our baseline model we used zero-shot-classification from transformers. We used the default model, facebook/bart-large-mnli, and received an accuracy of 38%.

```
# Initialize pipeline
classifier = pipeline('zero-shot-classification', model='facebook/bart-large-mnli', device=0)

# Define candidate labels
candidate_labels = ['Normal', 'Depression', 'Suicidal', 'Anxiety', 'Bipolar', 'Stress', 'Personality disorder']

# Run classification in batch
def classify(batch):
    results = classifier(batch['statement'], candidate_labels=candidate_labels)
    return {'status_label': [r['labels'][0] for r in results]}

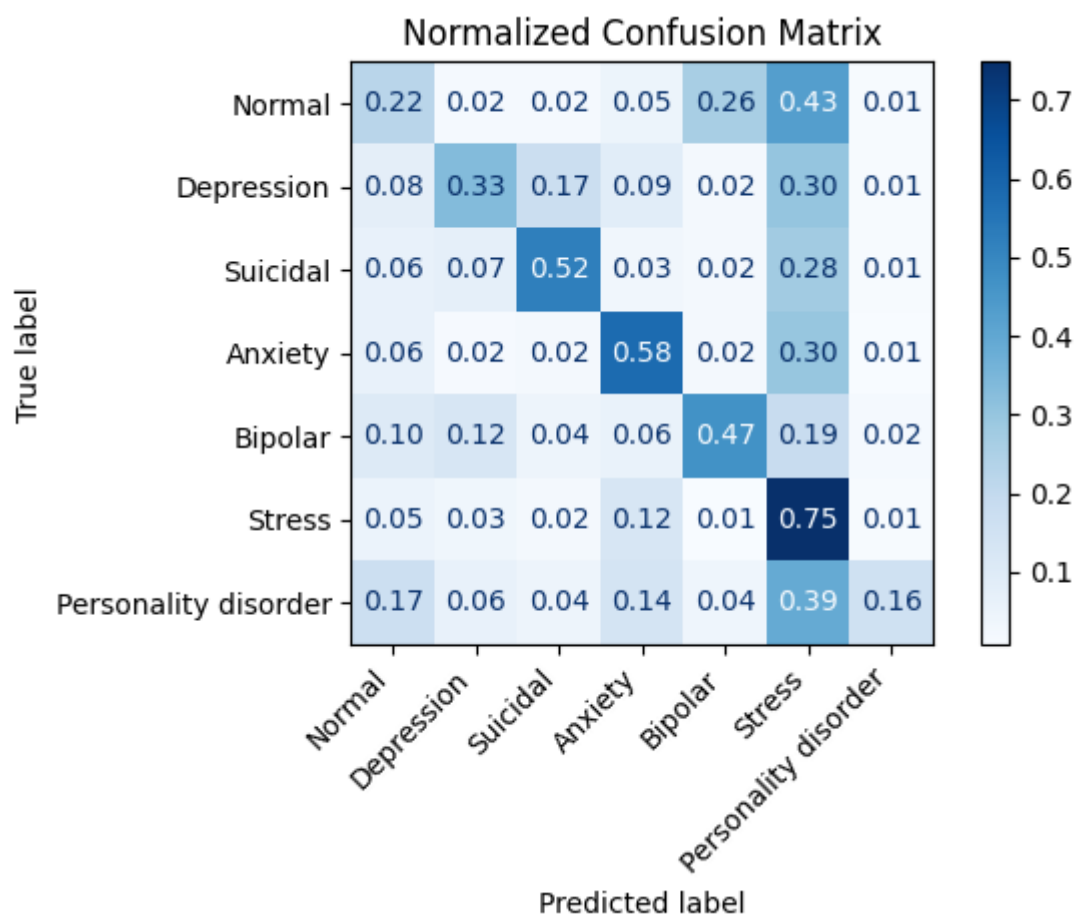
# Apply function
dataset = dataset.map(classify, batched=True, batch_size=16)

# Convert back to DataFrame
df = dataset.to_pandas()

# Evaluate
from sklearn.metrics import accuracy_score
print("Accuracy:", accuracy_score(df['status'], df['status_label']))
```

Not only did this model not output accurate results but we were also running into computational issues as this model had long runtimes which made it exceedingly difficult to try and assess multiple models and pre-processing techniques.

Figure 6



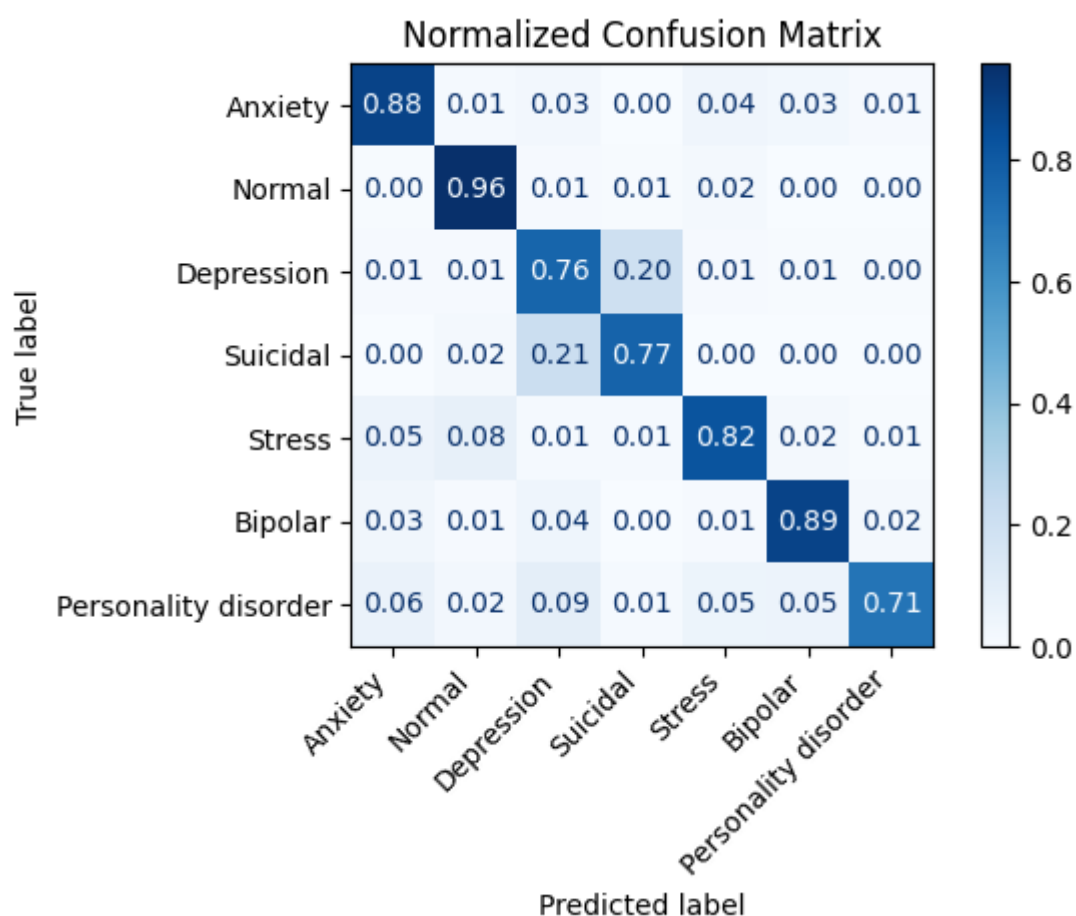
The above chart shows the confusion matrix from our zero-shot-classification testing. The results of this were similar for each model we evaluated. From the above chart we can see that this model predicted a majority of the statements to be Stress even though it is the second to last common label. This model also struggled to identify the Normal class showing it failed to identify differences between Normal and classes with a mental health label.

Next, we decided to look on Hugging Face to find models specifically trained for our use case. These models have already been trained on mental health related data, and we could further fine tune from there as needed. We decided to use the nateraw/bert-base-uncased-emotion model and got an accuracy score of 84%.

```
# Tokenizer from the mental health BERT model
model_name = "nateraw/bert-base-uncased-emotion"
tokenizer = AutoTokenizer.from_pretrained(model_name)
```

For our training arguments we used 4 epochs and a batch size of 16. We would have liked to spend more time tuning our model but due to the computation time of the model we were limited to the number of tests we could run.

Figure 7



The above chart shows the confusion matrix from our the nateraw/bert-base-uncased-emotion model. This shows a significant improvement over our baseline predicting across all mental health classes. The model had a slight problem differentiating between the Depression and Suicidal classes but did very well with the remaining classes. The model classified the Normal class the most accurately which shows the model was able to differentiate the Normal class the remaining classes with mental health problems with a low misclassification rate.

Conclusion, Implications, and Further Study

Mental health is growing concern in our country and across the world. Through our exploratory data analysis, we were able to point out the most common mental health

conditions in our data as well as common traits in the statements given by the individuals in our data.

Our baseline modeling did not give us favorable results in identifying mental health conditions in the text statements in our dataset. But we received much better results when we used a model pre-trained on mental health data and fine-tuned for our specific use case.

Mental health classification models like the one we built in this report can be used for early detection of signs of distress, depression, and anxiety from text that would allow for earlier professional intervention. Schools, workplaces, or healthcare systems are some potential industries that could benefit from a system like this to give students, employees, or patients support.

While our model performed well on the test data our performance on the validation set showed signs of potential overfitting. For further study on this project, we would consider doing more fine tuning to the parameters we used and test using other parameters to improve our performance on the validation set.

References

Dataset:

<https://www.kaggle.com/datasets/suchintikasarkar/sentiment-analysis-for-mental-health>¹

Research:

<https://www.nami.org/about-mental-illness/mental-health-by-the-numbers/>²